

DATA 505 (Statistics Using R): Final Exam

Part 2: Open Computer

Name: _____

2025-12-09

Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Problem 1: Two-Sample t-test (8 points)

We will analyze the `ToothGrowth` dataset, which contains data from an experiment on the effect of vitamin C on tooth growth in guinea pigs. The response is the length of odontoblasts (cells responsible for tooth growth) and one of the predictors is the delivery method (`supp`): orange juice (OJ) or ascorbic acid (VC).

```
# Load the data
data(ToothGrowth)
```

1 (2 points)

Use the `str()` function to examine the structure of the `ToothGrowth` dataset. How many observations are there?

```
# YOUR CODE HERE
```

Number of observations: WRITE YOUR ANSWER HERE.

2 (3 points)

Create side-by-side boxplots comparing tooth length (`len`) between the two supplement types (`supp`). You may use base R or `ggplot2`. Include appropriate axis labels.

```
# YOUR CODE HERE
```

3 (3 points)

Perform a two-sample t-test to determine if there is a significant difference in mean tooth length between the two supplement types. Use $\alpha = 0.05$. State your conclusion in context.

```
# YOUR CODE HERE
```

Conclusion: WRITE YOUR CONCLUSION HERE.

Problem 2: Multiple Linear Regression (9 points)

We will analyze the `swiss` dataset, which contains fertility and socio-economic indicators for 47 French-speaking Swiss provinces around 1888.

```
# Load the data
data(swiss)
```

1 (2 points)

Fit a simple linear regression model predicting `Fertility` from `Education` (percent of draftees with education beyond primary school). Store the model as `model1` and display the summary.

```
# YOUR CODE HERE
```

2 (2 points)

Now fit a multiple regression model predicting `Fertility` from `Education`, `Agriculture` (percent of males in agriculture), and `Catholic` (percent Catholic). Store this as `model2` and display the summary.

```
# YOUR CODE HERE
```

3 (2 points)

Compare the R-squared values of `model1` and `model2`. Does adding `Agriculture` and `Catholic` improve the model's explanatory power?

```
# YOUR CODE HERE
```

Answer: WRITE YOUR ANSWER HERE.

4 (3 points)

Use the `anova()` function to formally test whether `model2` is significantly better than `model1` at the $\alpha = 0.05$ level. What is your conclusion?

```
# YOUR CODE HERE
```

Conclusion: WRITE YOUR CONCLUSION HERE.

Problem 3: Logistic Regression (8 points)

We will analyze the `birthwt` dataset from the `MASS` package, which contains data on risk factors associated with low infant birth weight.

```
# Load the MASS package and data
library(MASS)
data(birthwt)

# The variable 'low' is our binary outcome:
# low = 1 means birth weight < 2500g (low birth weight)
# low = 0 means birth weight >= 2500g (normal)

# The variable 'smoke' indicates whether the mother smoked during pregnancy:
# smoke = 1 means the mother smoked
# smoke = 0 means the mother did not smoke
```

1 (3 points)

Fit a logistic regression model predicting `low` (low birth weight) from `smoke` (smoking status during pregnancy). Store the model as `smoke_model` and display the summary.

```
# YOUR CODE HERE
```

2 (2 points)

Based on the model summary, is there a statistically significant relationship between smoking and low birth weight at the $\alpha = 0.05$ level? Justify your answer.

Answer: WRITE YOUR ANSWER HERE.

3 (3 points)

Use the fitted model to predict the probability of low birth weight for two mothers: one who smoked during pregnancy (`smoke = 1`) and one who did not (`smoke = 0`). Report both probabilities.

```
# YOUR CODE HERE
```

Probability of low birth weight (smoker): WRITE YOUR ANSWER HERE.

Probability of low birth weight (non-smoker): WRITE YOUR ANSWER HERE.